1 hour

PAPER 2

Essay

Answer **four** *questions* **only.**

All questions carry equal marks.

All working must be clearly shown. Marks will **not** *be awarded for correct answers without corresponding working.*

1. (*a*) Given the sets A = {multiples of 3 less than 12}, B = {integers between 4 and 8} and C = {4, 5, 7}, find:
   (i) $A \cap B$;
   (ii) $(A \cup B) \cap C$;
   (iii) $(A \cap B) \cup C$.

   (*b*) Simplify: $1\frac{3}{4} - 2\frac{5}{6} - 1\frac{9}{10} + 4\frac{7}{8}$.

2. (*a*) Simplify: $15(4-6) \times 49 \div 7$.

   (*b*) Expand and simplify: $b(12a-3) - (a-b)(3+b)$.

   (*c*) Akosua walked for 3 hours at the rate of $1\frac{1}{2}$ km per hour from her village to Paamu to take a bus to Quamu. If the bus travelling at $15\frac{1}{2}$ km per hour takes 2 hours to travel from Paamu to Quamu,
   (i) what is the distance from Akosua's village to Quamu?
   (ii) how long would it take a man, riding a bicycle at 5 km per hour, to travel from Akosua's village to Quamu?

3. (*a*) (i) Express $8 \times 32 \times 4 \times 2$ in the form 2^m.
   (ii) Using your answer in (*a*)(i), state the value of m.

   (*b*) (i) Factorize the expression $\pi n^2 k - \frac{1}{4}\pi n^2 Q$.

   (ii) Use your answer in (*b*)(i) to find the value of the expression when $\pi = \frac{22}{7}$, $n = 2$, $k = 19$ and $Q = 20$.

   (*c*) Gifty and Justina shared an amount of GH¢ 418.00. If Gifty had 20 % more than Justina, how much did Justina receive?

4. (*a*) If $4m - 2(3 + 2m) + m(2m + 4) = 0$, find the values of m.

   (*b*) At a political rally, there were 240 women, 200 men, 160 boys and 120 girls.
   (i) Draw a pie chart to illustrate the information.
   (ii) What percentage of the people at the rally were females?

5. Madam Esi used $\frac{1}{4}$ and $\frac{2}{3}$ of her x acres of land to cultivate mangoes and oranges respectively.

(*a*) Express, in term of x, the number of acres of the land she used to cultivate:
(i) mangoes;
(ii) oranges.

(*b*) If madam Esi used 20 more acres of land to cultivate oranges than mangoes, find the value of x.

(*c*) How many acres of land was used to cultivate mangoes?

(*d*) Calculate, correct to the **nearest** whole number, the percentage of the land that was **not** used.

6. (*a*) Using a scale of 2 cm to 2 units on both axes, draw on a graph sheet two perpendicular axes, Ox and Oy, for the interval $-10 \leq x \leq 10$ and $-10 \leq y \leq 10$.

(*b*) On the same graph sheet, draw:
(i) a quadrilateral $ABCD$ with vertices $A(2, 4)$, $B(2, 8)$, $C(8, 8)$ and $D(8, 4)$;
(ii) the image $A_1B_1C_1D_1$ of $ABCD$ under a translation by vector $\begin{pmatrix} -5 \\ -2 \end{pmatrix}$, where $A \rightarrow A_1$, $B \rightarrow B_1$, $C \rightarrow C_1$ and $D \rightarrow D_1$;
(iii) the image $A_2B_2C_2D_2$ of $ABCD$ under a reflection in the y-axis, where $A \rightarrow A_2$, $B \rightarrow B_2$, $C \rightarrow C_2$ and $D \rightarrow D_2$.

(*c*) (i) What type of quadrilateral is $ABCD$?
(ii) Find the gradient of $\overline{A_2B_1}$.

END OF ESSAY TEST